

Extending the Spatio-Temporal Skew- t Process for Threshold Exceedance Prediction

AR(2) Temporal Priors and Multi-Resolution Spline Basis Functions

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July 2026

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Motivation

- **Threshold exceedance** events in air pollution (e.g., ozone) are spatially dependent and heavy-tailed.
- Gaussian processes cannot jointly capture **skewness** and **tail dependence**.
- This work extends the spatio-temporal skew- t process with:
 - ① AR(2) temporal priors, for richer temporal dynamics;
 - ② Multi-resolution spline basis functions, for non-stationary spatial structure.

Goal

Quantify the predictive gain of these extensions for **threshold exceedance prediction** (CRPS, Brier score).

Related Work

- skew- t processes and tail-dependence models.
- Temporal priors in spatio-temporal models.
- Basis-function expansions and autoFRK.

Model Setup

Let $Y(\mathbf{s}, t)$ be the observation at location $\mathbf{s}$ and time t :

$$Y(\mathbf{s}, t) = \mathbf{x}(\mathbf{s}, t)^\top \boldsymbol{\beta} + \mathbf{B}(\mathbf{s})^\top \boldsymbol{\omega}_t + \varepsilon(\mathbf{s}, t), \quad (1)$$

where

- $\mathbf{B}(\mathbf{s})$: multi-resolution spline basis functions;
- $\boldsymbol{\omega}_t$: temporal coefficients with an AR(2) prior

$$\boldsymbol{\omega}_t = \phi_1 \boldsymbol{\omega}_{t-1} + \phi_2 \boldsymbol{\omega}_{t-2} + \mathbf{e}_t;$$

- $\varepsilon(\mathbf{s}, t)$: skew- t error with skewness parameter $\boldsymbol{\lambda}$.

Inference & Evaluation

Bayesian inference

- MCMC sampling.
- Identifiability check for ϕ_T .

Predictive scoring

- CRPS.
- Brier score (threshold exceedance).

Simulation Design

Scenario	Parameters	Description
Strong	$(\phi_1, \phi_2) = (\cdot, \cdot)$	strong temporal dependence
Near-unit	$(0.15, 0.80)$	near unit root

Table 1: Simulation scenarios (illustrative).

- Key finding: the AR(2) advantage emerges in the near-unit scenario.

Simulation Results

(Place simulation result figures/tables here.)

Ozone Data Analysis

- Data source and preprocessing.
- After removing the autoFRK spatial structure, residuals become **heavier-tailed** (ν : $15.7 \rightarrow 3.8$).
- Heavy tails are a feature of the data, not a lack of fit.

Conclusion & Contributions

- ① AR(2) temporal priors yield significant predictive gains in the near-unit regime.
- ② Multi-resolution spline bases capture non-stationary spatial structure.
- ③ ϕ_τ is identifiable: the posterior concentrates well beyond the prior.

Future Work

Higher-dimensional space-time, more threshold settings, and real-time prediction.

References I

Thank You

Questions & Comments Welcome